



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Fifth Semester- Diploma in Engineering (Electrical Engineering) Session 2020-21

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Mid Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EE-5001	POWER SYSTEM OPERATION & PROTECTION	4	1	2	6	70	20	10	30	20	150
2	EE-5002	POWER ELECTRONICS & APPLICATION	4	1	2	6	70	20	10	30	20	150
3	EE-5003	INSTRUMENTATION	4	1	2	6	70	20	10	30	20	150
4	EE-5004	UTILIZATION OF ELECTRICAL POWER	4	1	2	6	70	20	10	30	20	150
5	EE-5005	ESTIMATING & COSTING	4	-	-	4	70	20	10	-	-	100
6	EE-5006	PROFESSIONAL ACTIVITIES	-	-	2	-	-	-	-	-	-	-
		Total	20	04	08	28	350	100	50	120	80	700

Theory Credits: 24
Practical Credits: 04
Total Credits: 28

Theory Marks: 350
Practical Marks: 120
Quiz, Mid Term, : 230

Total: 700



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

DIPLOMA IN ELECTRICAL ENGINEERING

(Courses effective from Academic Year 2019-20)

SYLLABUS OF COURSES TO BE OFFERED “DIPLOMA IN ENGINEERING”

Disclaimer: The graduate programme in Diploma in Engineering syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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EE-5001 POWER SYSTEM OPERATION AND PROTECTION

UNIT-I

Introduction to power system -Growth of power system in India, future schemes, various elements of power system. Interconnection, its necessity and advantages, planning, operation and improvement of power system. Load study, type of faults. AC/DC calculating board/digital computers.

Representation of power system -

Single line diagram, use of standard symbol. Per unit quantity - definition and advantages, base impedance conversion of per unit values from one base values to other base values. Generalized ABCD constants, their values in terms of circuit parameters. To prove $AD-BC=1$ characteristics of ABCD constants. The relations of Z_{so}, Z_{ro}, Z_{rs} in terms of generalized constants.

UNIT-II

Fault analysis -Type of faults - symmetrical and unsymmetrical. Phase sequence impedance phase sequence networks analysis of symmetrical faults. Analysis of L-G, L-L and L-L-G fault by symmetrical components, simple numerical problems.

Modern trends -E.H.T Lines, bundled conductors, HVDC system, block diagram, advantages and disadvantages. Block diagram of Power Line Carrier communication.

UNIT-III

Introduction to protection -Purpose of protective system, requirement and selection. Abnormalities in a power system and their effects. Reasons for failure. Self and non self clearing faults. Use of CT/PT in protective scheme. Definition of terms regarding CT/PT. Advantages of CT/PT. Neutral earthing, their methods and advantages.

Protective relay -

Type of relays- induction, electromagnetic, thermal. Primary and back up relaying. Types of back up relays, causes of failure of primary relaying. Explanation of terms used in relaying. Principle and working of different types of relays- electromagnetic and induction type. Induction type over current relay, reverse power relay, time and current settings. Differential relays, distance relays, thermal relays, inverse current characteristics.

UNIT-IV

Circuit interruption devices - Function of fuse. fusing factor, fusing characteristic. Application of isolator and circuit breakers Circuit breaker capacities Arc formation in C.B. and methods of arc extinction Definition of various terms with reference to circuit interruption waveform working principle and operation of Bulk Oil/Minimum oil/air blast/SFCB Merits and Demerits of different types of C.B.

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UNIT-V

Protection against over voltages - Causes and effects of over voltage.Traveling wave. Over-voltage protection, earth wire, lighting arresters- Multiple gap type, horn gap type, line type, station type and distribution type.Surge absorber.

Protective schemes -

Protection of alternator- various abnormalities, Merz price differential protection, over current and earth fault protection. Protection of transformer - various abnormalities, differential protection, Buchholz relay. Feeder and transmission line protection - time graded and over current protection, current graded system,differential protection. Protection of Induction motors use of thermal relays and under voltage protection.

LIST OF EXPERIMENTS:-

1. To determine the ABCD constants of a given N/T network.
2. To determine Z_{so} , Z_{ss} , Z_{ro} , Z_{rs} for a given N/ T network. Study of a HVDC system.
3. Study and use of CT/PT employed for protection and determine their ratio error.
4. Study of an Induction type O/C relay and plot the inverse characteristic.
5. To plot the inverse characteristic (current/time) of a given fuse wire. To determine the fusing factor of different fuse wire.
6. To study the differential protection of transformer/alternator.
7. Visit to power station
8. Study of different protective elements/schemes-ground wire
9. Lightning arrestors, fuse,circuit breaker. Study of Min. Oil Circuit breaker.

References Books:-

1. **Power electrics : Circuits, devices & applications, M. H. Rashid, PHI**
2. **Power Electronics, P.C. Jain, TMH**

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EE-5002 POWER ELECTRONICS & APPLICATION

UNIT-I

Thyristor -SCR - Structure, Two transistor model, characteristics, turn-on methods, factor responsible for temperature rise, circuit for over voltage, over current, voltage surge & high dv/dt, Gate production. Modes of heat transfer.

Triggering circuits - Resistance phase shift, UJT, Schmitt trigger. Turn-off types of commutation, methods of commutation. SCR rating, series and parallel connection. Thyristor family and other devices, DIAC, TRIAC, SCS, SUS, LASCR, MOSFET, IGRT, GTO, MCT.

UNIT-II

Rectification -Single phase HW and FW converters. Mathematical relations of I_{dc} and I_{rms} . Applications Advantages of polyphase rectification.

UNIT-III

Inverter -Series and parallel inverter, current and voltage source inverter, emergency tube light, comparison of inverters using different types of (transistor, MOSFET, IGRT) devices, block diagram of UPS (on load / off load). Applications of Inverter

UNIT-IV

Converter- Block diagram of DC to DC converter.Converter duty cycle, 4-quadrant operation of choppers,applicationsWorking of single phase cyclo- converter, basic concept of three phase cyclo-converter, application of cyclo-converters.

UNIT-V

Speed control of Motors -Advantages of electronic speed control, 4-quadrant operation of DC motors, constant torque and constant horse power operation, speed control of separately excited DC motor, single and 3 phase controlled rectifiers,field failure protection and armature current limiter, speed control using chopper, dual converter.Speed control of induction motor using rectifier, inverter and cyclo- convector (block diagram only).Speed control of slip ring induction motor using SCRS in rotor circuit.

Applications -

High frequency heating - induction and dielectric, principle, applications. Resistance welding, electronic line contactor, heat control circuit. Static AC & DC switch, over voltage protection, time delay circuits, reversing switch with plugging logic and digit circuits, PF corrections

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LIST OF EXPERIMENTS:-

1. Characteristics of thyristor family devices.
2. Study of single phase controlled rectifiers.
3. Study of emergency tube light
4. Study of SCR triggering circuits
5. Study of commutation circuits
6. Electronic speed control of d.c.motor
7. Electronic speed control of induction motor
8. Study of UPS
9. Study of chopper

References Books:-

1. Power electronics : Circuits, devices & applications, M. H. Rashid, PHI
2. Power Electronics, P.C. Jain, TMH

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EE- 5003 INSTRUMENTATION

UNIT-I

Measuring System Elements of a measuring system, Block diagram of system configuration, performance, standards, time lag, error, distortion and distortion meters, noise and noise factor.

UNIT-II

Transducers definition and classification, mechanical devices as primary detectors, Characteristic & choice of Transducers, Electrical transducers, Advantages of electric transducers, Active and passive transducers, Classification, Resistive, inductive and capacitive transducers, Potentiometric, Metallic and semiconductor strain gauges, Gauge factor, types, material used and applications. Thermistor, RTD, Inductive, LVDT, RVDT and Capacitive transducers and their application. Thermocouples, Piezo-Electric transducers, principle, materials used, mode of operation and application. Frequency generating transducers. Hall effect transducers, Opto-electronic transducers such as photo voltaic, Photo conductive, and photoconductive cells, constructional details, characteristics and applications. Photo diodes and transistors, characteristics and applications. Digital transducers, Optical encoders for linear and angular displacement measurement.

UNIT-III

Signal Conditioners Purpose of signal conditioning, Classification, Input modifier, Operational amplifiers circuits used in instrumentation, D.C. amplifier, chopper amplifier. Instrumentation amplifier, characteristics, three amplifier configuration. A/D and D/A converters.

UNIT-IV

Measurement of Physical Quantities: Measurement of Pressure-Types of pressure measurement devices, Force summing devices, Secondary transducers, Low pressure measurement, Pirani gauge and thermocouple gauge. Resistive, Inductive and Capacitive pressure measuring devices. Measurement of speed: Measurement of speed by stroboscope, photoelectric and reluctance pick-up devices for speed measurement. Measurement of Vibration-Necessity for Vibration Measurement, Seismic Transducer, Piezo-electric and LVDT Accelerometers. Measurement of Temperature-Temperature measuring devices, Resistance thermometers, Radiation and Optical Pyrometers. Measurement of Flow-Turbine and electromagnetic flow meters, Ultrasonic flow meter, Thermal flow meters. Measurement of Humidity-Humidity, absolute and relative humidity, Resistive and Capacitive hygrometers. Measurement of pH Value-Concept, pH scale, pH cell, pH meter. Measurement of Thermal Conductivity (gas analyzer). Measurement of level: Float and potentiometer method of flow measurement, Resistive and Capacitive methods, Ultrasonic and Gamma rays methods. Measurement of Force and Torque-

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Electronic weighing system, Hydraulic load cell. Torque measurement, concept, Stress and Deflection type torque measurement methods.

UNIT-V

Telemetry Necessity, Principle, classification, current and voltage telemetry, Position telemetry, synchros. Frequency and pulse telemetry, Principle of frequency and pulse modulation, PAM, PPM and PCM. Idea about landline and R.F. telemetry and multiplexing. Pulse code format, Modulation techniques of digital data transmission, Digital multiplexers.

Display Devices and Recorders: Digital display system and indicators like CRT, Seven Segment LED, LED, LCD. Analog and digital recorders, Strip and circular chart recorder and Magnetic tape recorder, X-Y recorders. Ultraviolet recorders, Frequency modulated (FM) recording. Digital tape recorders.

LIST OF EXPERIMENTS:-

1. Measurement of Distortion using wave distortion meter.
2. Measurement of load/weight using strain gauge and cantilever.
3. Measurement of linear displacement by LVDT and draw its characteristics.
4. Measurement of temperature by-
(a) Thermocouple (b) Resistance Thermometer
5. Measurement of pressure using LVDT and diaphragm gauge.
6. Study and use of data conversion using ADC and DAC.
7. Measurement of pH value using pH meter.
8. Measurement of Humidity by hygrometer.
9. Study and use of synchros in position telemetry system.
10. Measurement of Vibration using piezoelectric/LVDT transducer.
11. Study and flow measurement using electromagnetic flow meter.
12. Study of time division and frequency division multiplexing.
13. Measurement of liquid level by resistive/capacitive transducer.
14. Study and measurement of temperature using optical and radiation pyrometer.
15. Study of strip chart recorder and magnetic tape recorder.

REFERENCES:-

1. Electrical and Electronic Measurement and Instrumentation by A.K. Sawhney
2. Instrumentation Devices and Systems by C S Rangan, G R Sharma and V S V Mani
3. Digital Electronics By Malvino Leach
4. Instrumentation By Cooper

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EE-5004 UTILIZATION OF ELECTRICAL POWER

UNIT-I

Illumination-Electromagnetic wave spectrum, solid and plane angle, definition of electrical terms in use, sensitivity of human eye. Luminous efficiency, horizontal and vertical laws of illumination, definition of terms used in lighting, lighting scheme, various types of lamps, their use and fittings.

UNIT-II

Electric welding- Definition, classification of electrical welding, principle of arc welding. Qualities of a good weld. Welding defects. Resistance welding, advantages, classification, principle and working, comparison of resistance and arc welding process, A.C. & D.C. arc welding.

Power factor improvements- Causes of low P.F., effects of low P.F., methods of improvement of P.F. and its economics.

UNIT-III

Electric drives- Merits and demerits of electric drives, factors governing selection of motors, drive requirements. Group and individual drive, starting and running characteristics of various motors. Selection of starters, hand operated and contactor type starters, liquid resistor type starter. Speed control of motors, load equalization, use of fly wheel. Motor enclosures, selection of motors for particular service, size and rating of motors.

UNIT-IV

Electric heating- Advantages and disadvantages of electric heating, methods of electric heating. Principle of electric heating. Resistance heating, heating elements and alloys. Causes of failures of heating elements. Arc furnaces, principle, construction, working and uses. Induction heating principle, construction and use of Ajax Wyatt (core type) and coreless type. L.F. and H.F. induction furnaces. Dielectric heating principles and uses.

UNIT-V

Electro-chemical processes and storage batteries- Electro deposition and Faraday's laws of electrolysis, various electro-chemical processes like electroplating, electro-extraction, regions. Storage batteries, classification, construction. Battery maintenance, battery charging, circuit diagram. Application of storage batteries.

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LIST OF EXPERIMENTS:-

1. Speed control of slip ring induction motor by variation of rotor resistance.
2. To verify the change in power factor by changing load parameters and its improvement using capacitance.
3. To draw 'V' curves of synchronous motor.
4. Study and operation of resistance oven and to control its temperature. Study of dielectric / induction heating.
5. Measurement of luminous efficiency of lamps by lux meter. Study and operation of various types of lamps.
6. Study of arc welding.

REFERENCES:-

1. Utilization Of Electrical Power, K Prasad & Dr. H.M Rai

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UNIT-I

Elements of Estimating- Principles of estimating, purchase procedure, cost of materials, various charges like labor, stores, overhead tools, contingency etc.

Domestic and Industrial Wiring- Various types of wiring systems including P.V.C. pipe, their merits and demerits. Calculation of total load & selection of wire, preparation of estimates for a small residential building, big institution or office building. Estimate for single store yard, multistory building. Estimate for a small workshop and industrial installation, agricultural pump, domestic pump, flour mills etc. Estimation of total cost.

UNIT-II

Service connections- For a single storey and multistoried building, single phase and three phase service connections, various methods of service connections. Distribution of circuits for light and power load. (Guidance may be taken by the M.P. Electricity Board estimates).

UNIT-III

Substations- Various types of sub-stations, pole-mounted in-door and out-door substations. Estimating quantity and cost for a substation of a given specification.

UNIT-IV

Overhead lines : H.T. & H.T. lines Preparation of estimate and costing of 11KV or 33KV line. Selection of routes. Estimates for distribution lines- Location of poles for a given situation or locality. Providing street lights, necessary hardware, stay arrangements, underground cables, providing services lines using underground cables.

UNIT-V

Estimating and costing for repair/maintenance of electrical devices/equipments

Estimates for repairing electrical equipment e.g.

- Rewinding, assembling and testing of polyphase induction motor.
- Repairing of 3 phase starters.
- Repairing of single phase transformers.

Repairing of devices like hot plate, press, mixer fan etc.

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